Meta-Analysis of Venous Thromboembolism Prophylaxis in Medically Ill Patients

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ABSTRACT

Background: Venous thromboembolism (VTE) prophylaxis in medically ill patients has received a level 1A recommendation in previously published clinical guidelines. Pharmacologic prophylaxis for VTE includes unfractionated heparin (UFH), low-molecular-weight heparin (LMWH), and fondaparinux. Few direct comparisons between anticoagulants exist in medically ill patients.

Objective: This meta-analysis was conducted to assess UFH and LMWH (including the selective factor Xa inhibitor fondaparinux) in the reduction of in-hospital VTE in unselected medically ill patients.

Methods: We searched MEDLINE, EMBASE, and the Cochrane Controlled Trials Registry databases from January 1981 through September 2007 (English language) for randomized controlled trials using the following terms: *dalteparin*, *enoxaparin*, *fondaparinux*, *nadroparin*, and *heparin*. References of included articles and key review papers for additional studies were also searched. Data from studies were included in the analysis if the studies included medically ill patients with risk factors for VTE who had been followed up for 7 to 21 days.

Results: A total of 12,391 patients (of whom 8357 were in placebo-controlled trials) from 9 studies were included. Mean age for the entire cohort was 72.8 years; mean (SD) body mass index, 25.6 kg/m^2; and mean (SD) actual body weight, 68.2 kg. Deep vein thrombosis (DVT) was significantly reduced with the addition of an LMWH compared with placebo (odds ratio [OR], 0.60; 95% CI, 0.47–0.75; $P \leq 0.001$), but rates of DVT were similar when comparing LMWH with UFH (OR, 0.92; 95% CI, 0.56–1.52). No significant differences in pulmonary embolism (PE) or death were found among the UFH, LMWH, and placebo groups. LMWH was associated with a significant increased risk for minor bleeding compared with placebo (OR, 1.64; 95% CI, 1.18–2.29; $P = 0.003$). However, no significant difference was found between LMWH and UFH (OR, 0.68; 95% CI, 0.27–1.70). Major bleeding events were similar among all groups: LMWH/fondaparinux versus placebo, OR, 1.65 (95% CI, 0.8–3.4); LMWH/fondaparinux versus UFH, OR, 0.69 (95% CI, 0.29–1.68); LMWH/fondaparinux versus UFH or placebo, OR, 1.16 (95% CI, 0.66–2.04).

Conclusions: This analysis suggests that VTE prophylaxis with an LMWH (including fondaparinux) or UFH is effective in reducing the rate of DVT, but this benefit did not extend to enhanced protection against PE. Additionally, LMWH and UFH had similar bleeding outcomes. (*Clin Ther*. 2007;29:2395–2405) Copyright © 2007 Excerpta Medica, Inc.

Key words: venous thromboembolism, prophylaxis, low-molecular-weight heparin, unfractionated heparin, fondaparinux.

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INTRODUCTION

Venous thromboembolism (VTE), including deep vein thrombosis (DVT) and pulmonary embolism (PE), is a significant medical and financial burden in hospitalized patients. Historically, hospital-acquired DVT and PE are observed in 10% to 33% of all patients and up to 10% of medically ill patients,[1–8] which translates to increased health care costs with an inflation-adjusted treatment in hospital costs of VTE events ranging from US $12,760 to $18,557 per patient.[9] Moreover, recurrent VTE occurs in 35% of medically ill patients, of whom 23% will develop postthrombotic sydrome, making the long-term complications of VTE a major concern.[10,11] Consequently, guidelines from the American College of Chest Physicians provided a level 1A recommendation for primary prophylaxis of VTE in the medically ill population.[12] However, according to the DVT Free Registry, only 42% of inpatients receive appropriate prophylaxis, indicating an underutilization problem despite Agency for Healthcare Research and Quality recommendations.[13,14]

The pathophysiology of VTE is associated mainly with venous stasis and hypercoagulopathy. Major independent risk factors for VTE include paresis, acute infectious disease, congestive heart failure, malignancy, indwelling central venous catheter, varicose veins, and previous thrombosis.[15] Pharmacologic prophylaxis, aimed at improving venous flow and reducing coagulability, has been previously established with low-dose unfractionated heparin (UFH).[5,16,17] However, with the advent of low-molecular-weight heparin (LMWH), both UFH and LMWH are anecdotally considered therapeutically equivalent for the reduction of VTE in medically ill patients, but data to support similarity between UFH and LMWH have not yet emerged. A newer agent, the selective factor Xa inhibitor fondaparinux, was reported to have a 46.7% relative risk reduction of preventing VTE in medically ill patients when compared with placebo.[18] Although current (2004) guidelines do not make recommendations regarding the role of fondaparinux for the prevention of VTE in medically ill patients, this may evolve with the anticipated updated (2007) guidelines. Moreover, health care professionals recommend prophylaxis regimens based on outcomes data generated in placebo-controlled, rather than active-controlled, trials. There are limited controlled trials comparing anticoagulants for thromboprophylaxis in the general population of medically ill patients that provide outcomes data and guidance. Further investigation regarding the outcome benefits of LMWH agents as compared with UFH are thus warranted and should be explored. The objective of this meta-analysis was to systematically review randomized controlled trials (RCTs) of VTE prophylaxis in medically ill patients to determine whether there is a difference in efficacy and safety profiles between UFH, LMWH (or fondaparinux), and placebo.

MATERIALS AND METHODS

Study Identification and Selection

We systematically searched MEDLINE, EMBASE, and the Cochrane Controlled Trials Registry from January 1981 through September 2007 for English-language RCTs evaluating UFH, LMWH, or fondaparinux for the reduction of in-hospital VTE in the unselected medically ill population using the following terms: *dalteparin*, *enoxaparin*, *fondaparinux*, *nadroparin*, and *heparin*. We searched references of included articles and key review papers for additional studies, and studies were included in the analysis if they included medically ill patients with risk factors for VTE who had been followed up for 7 to 21 days. We clustered dalteparin, enoxaparin, nadroparin, and fondaparinux together in the LMWH/fondaparinux group. Fondaparinux was included with LMWH agents for practicality since fondaparinux has been regarded as an "ultra-LMWH" and because the pharmacokinetic and pharmacodynamic properties of fondaparinux are similar to other LMWH agents,[19] although the US Food and Drug Administration has recently classified it as a selective factor Xa inhibitor.[20,21] Efficacy end points of any VTE, DVT, or fatal or nonfatal PE (considered thrombotic events) were selected for comparison, as well as tolerability end points including major or minor bleeding, fatal bleeding, and VTE-related death. We used the diagnostic criteria for *medically ill*, *VTE*, *DVT*, *PE*, *major bleeding*, and *minor bleeding* provided by each study. Studies failing to report drug doses or excluding end points of VTE, fatal or nonfatal PE, or bleeding were excluded from the analysis.

Study Quality

Three reviewers (A.O.K., M.A.S., and J.L.D.) independently evaluated each study for inclusion and exclusion criteria and tolerability and efficacy end points and assigned Jadad scores.[22,23] We only included RCTs with a corresponding Jadad score of 3 or

greater (maximum score, 5). Discrepancies in Jadad score assignment and inclusion criteria were resolved through consensus and discussion. We abstracted study characteristics including number of patients, patient age, sex, reason for admission, risk factors for VTE events, drug, dose, route, schedule, efficacy end points (VTE, DVT, and PE) and tolerability end points (major and minor bleeding).

Data Abstraction and Statistical Analysis

As the purpose of this analysis was to compare the incidence of death, thromboembolic events, and bleeding events in medically ill patients receiving UFH, LMWH/fondaparinux, or placebo, we extracted from each study data to create 3 major comparisons of efficacy evaluating LMWH/fondaparinux against placebo, LMWH/fondaparinux against UFH, or the combination of LMWH/fondaparinux versus UFH or placebo for the reduction of VTE and the occurrence of new bleeding. Data pertaining to each event were represented as proportions and a meta-analysis was constructed using Comprehensive Meta Analysis version 2.2 (Englewood, New Jersey). We also used the Der Simonian and Laird random-effects model to calculate pooled odds ratios (ORs) and corresponding 95% CI, with a P value ≤ 0.05 considered statistically significant.[24] The I2 statistic was used to estimate variability from heterogeneity in the effect size of each selected end point.[25] Meta-regression, according to baseline risk factors, was performed to explore additional relationships between efficacy and tolerability end points. The statistical analysis was performed by one reviewer (M.A.S.), and all authors had access to the statistical database, evaluated the results, and agreed on the data submitted for publication.

RESULTS

Literature Search

The initial search identified 881 articles, which have been summarized in **Figure 1**. After examining the abstracts and titles, we selected 14 trials for more detailed evaluation, of which 9 fulfilled the predefined inclusion criteria (**Table I**).[18,26–33] All included reports were RCTs. The 5 excluded trials included patients who failed to achieve a minimum Jadad score of 3, mainly due to lack of randomization.[5,16,17,34,35] Although patients who have experienced stroke are regarded as medically ill, trials that comprised solely stroke patients were excluded as a recently published meta-analysis addresses VTE prophylaxis in this population, and the objective of the current study was to examine an unselected medically ill population as seen in routine clinical practice.[36]

Demographic Characteristics

The total number of patients included in this analysis was 12,391, of which 8357 were in placebo-controlled trials.[18,26–29] The mean age for the entire cohort was 72.8 years, with similar distributions in regard to other baseline characteristics (**Tables I** and **II**). Mean body mass index was 25.6 kg/m^2 and mean actual body weight was 68.2 kg. Heart failure was reported more frequently (29.4%) than acute respiratory disease (22.4%) or infection/inflammation (24.7%) (**Table II**). The majority of trials excluded patients who were mechanically ventilated, had active bleeding, renal failure, conditions requiring full anticoagulation, or who were pregnant.

Thrombotic Events

DVT events were significantly reduced with the addition of LMWH/fondaparinux compared with placebo (OR, 0.60; 95% CI, 0.47–0.75; $P \leq 0.001$) (**Figure 2**) and were associated with an absolute risk reduction (ARR) of 1.36% and a number needed to treat (NNT) of 74. Of note, this reduction was driven by dalte-

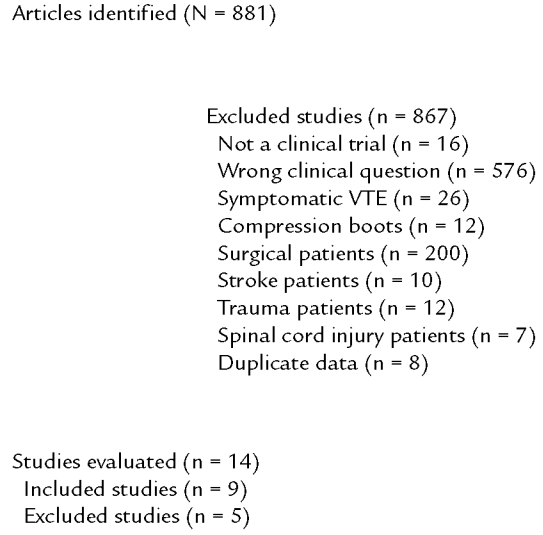

Figure 1. Results of the literature search. VTE = venous thromboembolism.

Table I. Characteristics of trials included in the meta-analysis.

Trial	Antithrombin	Dose and Duration	Control (Active/Placebo)	Primary End Point	No. of Patients	Year
Samama et al[26]	Enoxaparin	20 or 40 mg/d for 6–14 d	Placebo	DVT/PE/VTE at day 14	1102	1999
Mahe et al[29]	Nadroparin	7500 U/d for 21 d	Placebo	Mortality at 21 days	2477	2005
Fraisse et al[27]	Nadroparin	3800–5700 U/d for 11 d	Placebo	DVT/VTE at day 11	223	2000
Leizorovicz et al[28]	Dalteparin	5000 U/d for 14 d	Placebo	DVT/PE/VTE/ sudden death at day 21	3706	2004
Cohen et al[18]	Fondaparinux	2.5 mg/d for 6–14 d	Placebo	DVT/VTE at day 15	849	2006
Bergmann and Neuhart[30]	Enoxaparin	20 mg/d for 10 d	Heparin 5000 U BID	DVT/PE/VTE at day 10	442	1996
Harenberg et al[32]	Enoxaparin	36 mg TID for 10 d	Heparin 5000 U BID	DVT/PE/VTE at day 10	1968	1996
Lechler et al[31]	Enoxaparin	40 mg/d for 7 d	Heparin 5000 U BID	DVT/PE/VTE at day 7	959	1996
Kleber et al[33]	Enoxaparin	40 mg/d for 10 d	Heparin 5000 U BID	DVT/PE/VTE at day 10	665	2003

DVT = deep vein thrombosis; PE = pulmonary embolism; VTE = venous thromboembolism.

Table II. Demographic characteristics of patients included in the meta-analysis.

Reason for Admission	No. (%) of Patients*	Major Risk Factors for VTE	No. (%) of Patients*
Congestive heart failure	3644 (29.4)	Elderly	3435 (27.7)
Acute respiratory disease	2778 (22.4)	Previous VTE	857 (6.9)
Acute infection or inflammation	3066 (24.7)	Current or previous cancer	827 (6.7)
Multiple reasons	256 (2.1)		

VTE = venous thromboembolism.
*Of 12,391 total patients.

parin evaluation as reported by Leizorovicz et al.[28] The remaining 4 LMWH/fondaparinux studies did not find an association with reduced events compared with placebo at 7 to 21 days.[18,26,27,29] No significant difference was found in the incidence of DVT when comparing LMWH/fondaparinux with UFH alone (OR, 0.92; 95% CI, 0.56–1.52), suggesting that LMWH/fondaparinux and UFH are similar in reducing DVT events in medically ill patients (**Figure 3**). However, when LMWH/fondaparinux was compared with the combination of UFH or placebo, a significant reduction of DVT events was observed (OR, 0.64; 95% CI, 0.51–0.79; $P \leq 0.001$), and these data were associated with an ARR of 2.1% and an NNT of 48 (**Figure 4**).

	OR	Lower Limit	Upper Limit	n/N LMWH	Placebo	n/N *P*
DVT	0.60	0.47	0.75	135/4017	171/3622	≤0.001
PE	0.54	0.28	1.05	15/4017	30/3622	0.07
VTE	0.59	0.47	0.74	150/4017	196/3622	≤0.001
VTE death	0.96	0.40	2.32	68/4017	68/3622	0.94
Any bleeding	1.68	1.24	2.28	67/2378	151/2786	0.001
VTE death or any bleeding	1.35	1.07	1.69	219/4017	135/3622	0.01

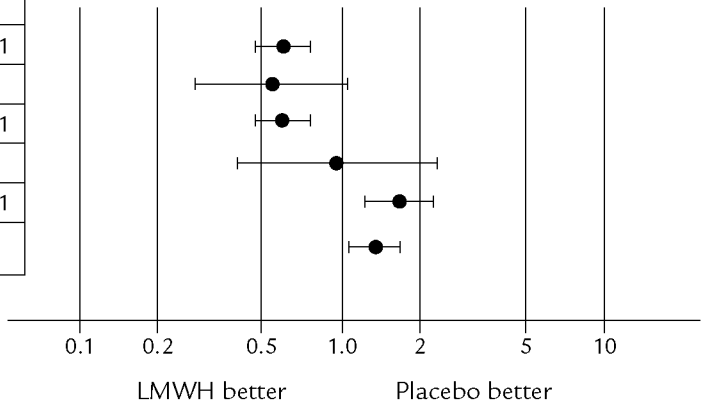

Figure 2. Low-molecular-weight heparin (LMWH)/fondaparinux versus placebo. OR = odds ratio; DVT = deep venous thrombosis; PE = pulmonary embolism; VTE = venous thromboembolism.

	OR	Lower Limit	Upper Limit	n/N LMWH	n/N UFH	n/N *P*
DVT	0.92	0.56	1.52	31/1771	34/1811	0.75
PE	0.80	0.22	2.89	4/1771	7/1811	0.73
VTE	0.89	0.54	1.46	32/1770	38/1811	0.64
VTE death	0.76	0.36	1.6	13/1806	17/1845	0.47
Any bleeding	0.72	0.44	1.18	28/1806	40/1850	0.19
VTE death or any bleeding	0.73	0.48	1.10	41/1806	57/1850	0.13

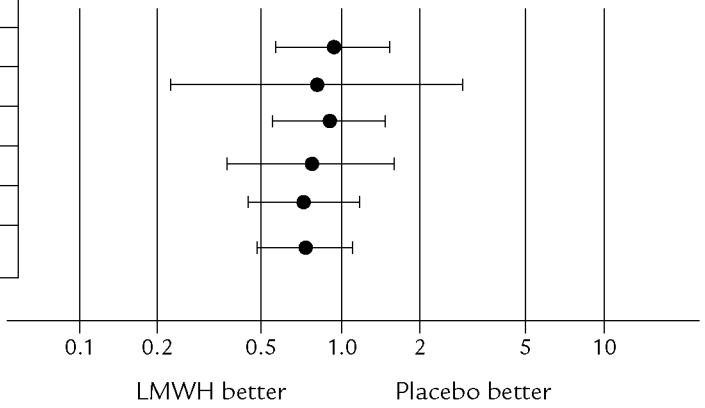

Figure 3. Low-molecular-weight heparin (LMWH)/fondaparinux versus unfractionated heparin (UFH). OR = odds ratio; DVT = deep venous thrombosis; PE = pulmonary embolism; VTE = venous thromboembolism.

	OR	Lower Limit	Upper Limit	n/N LMWH	n/N UFH or Placebo	n/N *P*
DVT	0.64	0.51	0.79	166/5788	209/5433	≤0.001
PE	0.59	0.34	1.03	18/5788	37/5433	0.065
VTE	0.64	0.52	0.79	185/5788	241/5433	≤0.001
VTE death	0.97	0.65	1.45	81/5822	85/5467	0.87
Any bleeding	1.25	0.87	1.8	179/4592	107/4228	0.23
VTE death or any bleeding	1.15	0.88	1.5	260/5822	192/5472	0.31

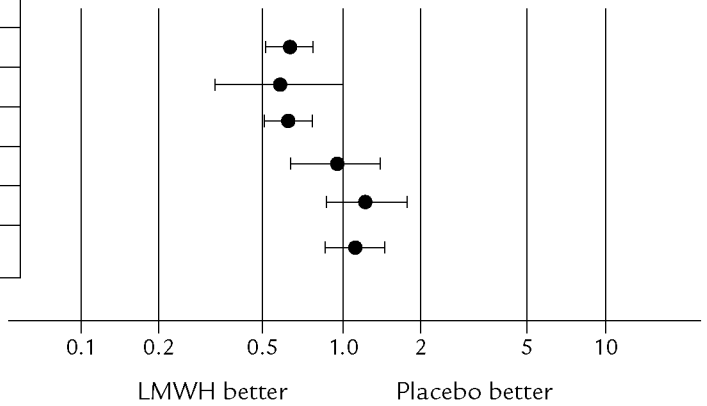

Figure 4. Low-molecular-weight heparin (LMWH)/fondaparinux versus unfractionated heparin (UFH) or placebo. OR = odds ratio; DVT = deep venous thrombosis; PE = pulmonary embolism; VTE = venous thromboembolism.

A reduction in PE events was not found when LMWH/fondaparinux was compared with placebo (OR, 0.54; 95% CI, 0.28–1.05). This finding remained consistent when LMWH/fondaparinux was compared with UFH (OR, 0.8; 95% CI, 0.22–2.9) and when LMWH/fondaparinux was compared with UFH or placebo (OR, 0.59; 95% CI, 0.34–1.03).

LMWH/fondaparinux was shown to significantly reduce VTE (DVT or PE) when compared with placebo (OR, 0.59; 95% CI, 0.47–0.74; $P < 0.001$) with an ARR of 1.68% and an NNT of 60, and when compared with UFH or placebo (OR, 0.64; 95% CI, 0.52–0.79; $P < 0.001$); the ARR was 1.15% and the NNT was 87. However, no significant difference was found between LMWH or UFH in reducing the incidence of VTE (OR, 0.89; 95% CI, 0.54–1.46).

No significant differences in VTE-related death were found between any of the comparative groups: LMWH/fondaparinux versus placebo, OR, 0.96 (95% CI, 0.40–2.32); LMWH/fondaparinux versus UFH, OR, 0.76 (95% CI, 0.36–1.6); or LMWH/fondaparinux versus UFH or placebo, OR, 0.97 (95% CI, 0.65–1.45). Of note, only 2 LMWH/fondaparinux studies reported VTE-related death as an outcome[18,28] and 3 studies[31–33] reported this outcome in UFH studies.

Bleeding

LMWH/fondaparinux was associated with a significantly increased risk for minor bleed compared with placebo (OR, 1.64; 95% CI, 1.18–2.29; $P = 0.003$), with an absolute risk increase (ARI) of 2.24% and a number needed to harm (NNH) of 45. Of note, this increased risk was driven by one evaluation of enoxaparin.[26] There was no significant difference in the incidence of minor bleeding between LMWH/fondaparinux and UFH (OR, 0.68; 95% CI, 0.27–1.70) or between LMWH/fondaparinux and UFH or placebo (OR, 1.3; 95% CI, 0.86–1.97). Major bleeding events were similar among all comparative groups: LMWH/fondaparinux versus placebo (OR, 1.65; 95% CI, 0.8–3.4); LMWH/fondaparinux versus UFH (OR, 0.69; 95% CI, 0.29–1.68); LMWH/fondaparinux versus UFH or placebo (OR, 1.16; 95% CI, 0.66–2.04). When minor and major bleeding events were combined, a significant increase in the incidence of any bleeding was shown when comparing LMWH/fondaparinux with placebo (OR, 1.68; 95% CI, 1.24–2.27; $P \leq 0.001$). Of note, this increased risk of any bleeding with an LMWH/fondaparinux was driven mainly by a trial of dalteparin and enoxaparin.[26,28] No significant difference was observed when comparing LMWH/fondaparinux with UFH (OR, 0.72; 95% CI, 0.44–1.18) or LMWH/fondaparinux with UFH or placebo (OR, 1.25; 95% CI, 0.87–1.80).

Any Bleeding or VTE Death (Net Benefit)

The composite end point of any bleeding or death from VTE was also significantly increased when comparing LMWH/fondaparinux with placebo (OR, 1.35; 95% CI, 1.07–1.70; $P = 0.01$), with an ARI of 1.73% and an NNH of 58, which was driven by an increase in minor bleeding. This difference was not observed when comparing LMWH/fondaparinux with UFH (OR, 0.73; 95% CI, 0.48–1.32), or LMWH/fondaparinux with UFH or placebo (OR, 1.15; 95% CI, 0.88–1.50).

DISCUSSION

LMWH/fondaparinux was associated with a significant reduction in the incidence of DVT compared with placebo in medically ill patients at days 7 through 21. No significant differences were found between LMWH/fondaparinux and UFH. However, the rate of PE was similar between LMWH, UFH, and placebo. These data suggest that pharmacologic prophylaxis with LMWH or UFH is effective in preventing DVT, but this benefit did not extend to reduction in PE, VTE, or death. Furthermore, no significant differences were observed in minor, major, or composite bleeding between LMWH and UFH; therefore, when evaluating both thrombotic and bleeding measures, LMWH and UFH were found to have similar safety profiles and efficacy in these medically ill patients between 7 and 21 days.

Cursory evaluations of these data may suggest that LMWH/fondaparinux may provide better protection against DVT since there was an early reduction in DVT events in patients treated with LMWH/fondaparinux compared with placebo between 7 and 21 days, based on data from dalteparin studies. This effect was inconsistent, however, as there was no significant difference in DVT between enoxaparin, fondaparinux, or nadroparin or versus placebo at this time point. Yet, at 30 days enoxaparin, fondaparinux, and nadroparin each was associated with a reduction in DVT compared with placebo.[18,26–33] These findings may be attributed to the duration of study and to the number of patients receiving dalteparin compared with the other

LMWH/fondaparinux agents: ~44% of patients in the LMWH/fondaparinux group received dalteparin, which may have led to the overestimation of the prophylaxis effect of this agent in general.[28] The reduction in the incidence of VTE (DVT and PE) was observed in the LMWH/fondaparinux group compared with the placebo group, mainly due to the reduction of DVT driven by dalteparin. When we compared LMWH/fondaparinux to UFH, there were no significant differences in thrombotic events. Despite a pooling of similar studies, this meta-analysis remained underpowered to detect a significant reduction in PE with LMWH prophylaxis versus placebo through 21 days despite having >8300 patients in the placebo comparison and >12,000 in the combined analysis.

As expected, we observed more minor bleeding events with LMWH/fondaparinux than placebo, but similar rates between LMWH/fondaparinux and UFH. Comparison suggests that treating 150 patients with LMWH/fondaparinux would prevent 2 DVT events and may result in 3 instances of minor bleeding. This comparison suggests that the ratio of benefit/harm favors the routine use of pharmacologic prophylaxis for DVT in the unselected, hospitalized, medically ill population as indicated in current guidelines.

Limitations to this meta-analysis should be discussed, including differences across and within studies. First, not all studies used the same criteria for VTE diagnosis or bleeding definitions. In addition, it may not be ideal to group all LMWHs in 1 drug class because there are subtle differences between LMWH agents. We included fondaparinux in the LMWH group because (1) it has similar pharmacokinetic and pharmacodynamic characteristics to other LMWH agents; (2) it is the only selective factor Xa inhibitor available for comparison; and (3) only 1 trial of fondaparinux in the medically ill population is available for comparison. Recognizing these limitations, and in the absence of other comparator trials, we felt it was reasonable to combine fondaparinux with other LMWH agents into 1 group. Also, because the focus of this analysis was to look at outcomes, the issue of strict pharmacologic classification was judged to be less important than the analysis of existing evidence of efficacy or safety profiles in the routine clinical practice. Additionally, only English-language studies were included in this meta-analysis. RCTs comparing LMWH agents and fondaparinux are the next step in determining whether specific agents are clearly associated with a reduction in thrombotic events in the medically ill population.

CONCLUSIONS

The rates of VTEs appear to be reduced in medically ill patients when addressed with appropriate pharmacologic prophylaxis. This analysis suggests that VTE prophylaxis with an LMWH (dalteparin, enoxaparin, fondaparinux, or nadroparin) or with UFH is effective in reducing the rate of DVT, but this benefit did not extend to enhanced protection against PE. Additionally, LMWH agents and UFH had similar bleeding outcomes. An additional consideration is that dalteparin, enoxaparin, and nadroparin are associated with heparin-induced thrombocytopenia, whereas fondaparinux is not. Therefore, it is important to consider patient- and institution-specific factors when selecting pharmacologic prophylaxis (ie, renal function, heparin-induced thrombocytopemia risk, and formulary restrictions).

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